TRACK C: CONSUMER, FMCG & RETAIL

Team: **Who cares** | Author: **Abhay Singh & Team**

ARGUS: Autonomous Financial Discovery Agent

ARGUS autonomously investigates public filings, tests competing explanations, and produces auditable financial discoveries.

Deliverables: Live Streamlit App · GitHub Repository · 2-Minute Video Demo

1. IDEA & APPROACH: WHY TRUE DISCOVERIES BEAT BASIC AI SUMMARIES — CORE INSIGHT

Most business AI tools today are just **summarizers**: you paste a 30-page company speech, and the AI repeats whatever good news the CEO said. But company leaders are paid to sound positive. The real business story is almost always hidden in the gap between what executives *say* and what their audited accounts and store numbers *actually show*.

ARGUS is not a chatbot or a summarizer. It operates like an independent investigative auditor through six simple actions: **Discover** (spot unusual gaps), **Connect** (link executive words with official balance sheets), **Calculate** (run exact math with no guesswork), **Reason** (test whether growth is genuine), **Challenge** (play devil's advocate to find fair explanations), and **Prove** (deliver clear findings with clickable public links).

> **Strict Rules:** Zero stock tips or investment recommendations (no "buy", "sell", "hold", or price targets). 100% public information only—nothing paid or behind a password.

2. TOOLS & STACK: REAL TECHNOLOGY (NOT JUST VAGUE "AI") — ARCHITECTURE

Tool / Component	What Powers It	Why It Was Chosen
Public Filings Harvester	Screener.in Financial Tables & BSE India PDF Announcements	100% free, public records of audited quarterly P&L, balance sheet schedules, and investor call transcripts.
Autonomous Agent Framework	Python 3 Pipeline (Urllib, HTML Parsers, RegEx Engine)	Automates source collection, deterministic calculations and evidence-based hypothesis testing, with explicit handling of missing or conflicting data.
Deterministic Math Engine	Standard Python Arithmetic Module (No LLM Math)	Ensures exact calculations for growth rates, DSO, and turnover days without relying on language model math guesswork.
Reasoning & Red-Teaming	Gemini API (Structured Analytical Prompts)	Formulates investigative questions and stress-tests findings against fair industry counter-arguments.
Single Output Interface	Interactive Streamlit Web App & Altair Visual Engine	Clean, responsive single-screen dashboard with direct public regulatory links, deterministic math charts, and export tools.

3. WORKFLOW: FROM COMPANY NAME TO VERIFIED DISCOVERY — 5-STEP PROCESS

1. Company — Select or enter ticker (e.g. HINDUNILVR, TRENT) → **2. Investigation** — Pick weak spots: Price hikes vs Volume? → **3. Research Plan** — Harvest 12 quarters of P&L, notes & calls → **4. Evidence** — Cross-examine sources: CEO + Math + Rivals → **5. Discovery** — 1-sentence finding + exact disproof rule

How it thinks: The agent does not read aimlessly. It immediately asks: *"Are higher sales coming from selling more goods, or did they just raise prices? Are distributors taking longer to pay their bills? Are new store openings hiding slowing sales at older stores?"*

4. TRACK & GENERALITY: ELIGIBLE TRACK C UNIVERSE — COMPANY-AGNOSTIC RESEARCH

The research pipeline uses company-agnostic analysis rules for eligible NIFTY 50 companies in Track C. It can extend to other consumer-sector peers where permitted and where comparable public data is available:

A. Factory Brands (FMCG Makers)

E.g., Hindustan Unilever, Britannia, Dabur, Nestle, Bikaji (outside NIFTY 50).

Checks if higher revenue is just from raising prices while unit volumes are sluggish, or if warehouse stock is piling up as distributors take longer to clear products.

B. Retail Stores & Jewellery (Non-Volume Disclosers)

E.g., Trent (Westside/Zudio), Titan (Jewellery/Watches), V2 Retail, DMart.

Retailers do not report volume in kilograms. ARGUS tracks whether opening new stores creates an illusion of growth while older stores experience maturing sales throughput.

Cross-Examination, Self-Falsification & The Working Dashboard

5. TRIANGULATION: HOW CLAIMS ARE FACT-CHECKED & PUBLIC DATA PROVENANCE 100% PUBLIC LEDGER

Claims are cross-checked against independent public sources and official regulatory filings. Zero paywalls or private logins; every data point is auditable via free public links:

Evidence Leg	Public Source & Direct Ledger Link	How a Contradiction is Caught
Leg 1: Leadership Story	**BSE India Regulatory Filings** bseindia.com/stockinfo/AnnPdfOpen.aspx ↗	The CEO announces: *"Demand accelerating across retail."* But concall text reveals promotional discounting & store payback delays.
Leg 2: Audited Math	**Screener.in Audited Statements** screener.in/company/[TICKER]/consolidated/ ↗	Balance sheet schedules show network inventory grew +68.2% (₹2,289 Cr), locking up working capital in unsold inventory.
Leg 3: Industry Rivals	**Sub-Industry Peer Disclosures** screener.in/company/[PEER]/consolidated/ ↗	Competitor disclosures show steady inventory velocity, proving the inventory stretch is company-specific rather than a macro slowdown.

6. SELF-FALSIFICATION: THE "CHANGE-MY-MIND" RULE COUNTERARGUMENT & DISPROOF

ARGUS actively guards against confirmation bias through two built-in algorithmic sanity checks:

1. The Fair Alternative (Devil's Advocate)

The agent actively tests honest operational explanations: *"Did inventory rise because management bought raw materials ahead of inflation, or did new stores require baseline stock?"*

2. The Exact Disproof Condition (Measurable)

The system sets an exact milestone that disproves its concern: *"Disproven if the company delivers **two consecutive quarters** of verified volume growth >+4.0% with unpaid bills shrinking."*

7. OUTPUT FORMAT: SINGLE INTERACTIVE DASHBOARD (THE 8 ELEMENTS) INTERACTIVE DASHBOARD

Standard 8-Element Output: (1) 1-Sentence Discovery • (2) Why It Matters (4 bullets) • (3) Key Numbers & Altair Charts • (4) Multi-Quarter Timeline • (5) Triangulated Evidence Trail with Direct Public Links • (6) Devil's Advocate Counterargument • (7) Measurable Disproof Milestone • (8) Evidence Strength Rating.

ARGUS DISCOVERY DASHBOARD • Track C: Consumer, FMCG & Retail [Open Live Dashboard ↗] • [Export Memo PDF]

Pipeline: 🔍 Collect filings → 🧱 Calculate metrics → 🛡️ Cross-check → ✨ Publish discovery ⚡ Run Investigation

Company: HUL / Trent • **Sector Model:** FMCG / Retail • **Quarters Loaded:** 13 • **Sources Found:** 27

📈 Revenue Growth: +16.9% 📦 Inventory: 61.2 days (+68.2%) ⏱️ DSO: 0.9 days

1. THE DISCOVERY: Trent Ltd reports headline revenue expansion (+16.9% YoY), yet growth is driven by new store additions while network inventory expanded +68.2% (reaching ₹2,289 Cr), indicating that top-line acceleration masks maturing sales productivity at older locations.

2. WHY IT MATTERS (4 Bullets)
- 📈 **Growth:** New stores mask older store sales slowdown
- 📦 **Inventory:** Network inventory up +68.2% (₹2,289 Cr working capital)
- 💰 **Pricing:** Lease liabilities & markdown risk if velocity cools
- 🏬 **Distribution:** Expansion in Tier 2/3 vs. store payback

3. NUMBERS & CHARTS + 4. TIMELINE

[Growth: +16.9%] [Stock Days: 61.2d] [DSO: 0.9d]
[Chronological Altair Chart: Q1 ₹4,818Cr • Q4 ₹5,755Cr]
Timeline: Jun 2023 baseline → Mar 2026 expansion

5. EVIDENCE TRAIL (Direct Public Links)

[Audited Results]: Screener.in Ledger ↗
[Earnings Call]: BSE Concall Transcript PDF ↗
[Competitors]: Peer Benchmarks ↗

6. COUNTERARGUMENT & 7. FALSIFICATION

Counter: New stores take 18-24 months to mature.
Disproof Rule: Two consecutive quarters of sales growth >8.0% with stock turnover under 50 days.

8. ROADMAP: BUILD PLAN & MILESTONES TO 25 OCT BUILD PLAN & MILESTONES

Date / Milestone	Status: Built vs. Planned	What It Delivers for the Finals
Oct 19, 2026 (Today) **Working MVP**	**BUILT & TESTED** Autonomous execution.	Deployed live at **Streamlit Cloud ↗**. Pulls live filings, runs deterministic math, detects retail/FMCG divergence, and renders the 8-element dashboard.
Oct 20–21 **Concall Deep Dive**	**IN PROGRESS**	Extracts exact spoken sentences and speaker names from full-length concall PDFs using fast local text matching.
Oct 22–23 **Peer Matrix**	**PLANNED**	Broadens automated cross-checks from 3 peers to 10+ peers across non-NIFTY mid-caps and regional competitors.
Oct 24–25 **Finals Ready**	**PLANNED**	Stress-tested on 50+ Indian consumer tickers to deliver a flawless live 10-minute presentation on whichever company the judges pick.